

Review



Computational efficiency and accuracy of the Neighbored Element Method

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ABSTRACT

Gradient-enhanced regularization is a frequently utilized method for addressing numerical issues in material modeling. As a consequence of the regularization scheme, Laplacian terms will emerge in the strong form of evolution equations for additional field variables, also called internal variables. In a series of previous works, the Neighbored Element Method (NEM) was presented as a combination of the Finite Element Method and a generalized finite difference scheme with a weighted least-square method to approximate the Laplacian. The objective is the efficient solution of the total system of equations containing Laplacian and gradient terms. The systems of equations used with the NEM have a similar structure to, e.g., the heat and diffusion equation. In this study, the NEM is evaluated even further, in comparison to a well-established FEM routine with respect to accuracy and computational efficiency through investigating a chemo-thermo-mechanical system. It is demonstrated that the accuracy improves with a reduction in element size and, for an appropriate mesh, the relative average error is below 1%. These highly accurate solutions can be achieved with a notable reduction in computational time and memory cost of up two orders of magnitude, for approximately 620000 nodes with five degrees of freedom per node. This new technique can be applied to arbitrary solid finite element types and/or irregular meshes.

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1. Introduction

In many problems of continuum mechanics, additional degrees of freedom are introduced to describe the behavior of various multi-physical problems for regularized formulations. Examples can be found in topology optimization, damage mechanics [1–8], phase transformation and bio-mechanical problems [9–11]. Regularization of the additional degrees of freedom is often unavoidable due to the ill-posed mathematical fundamental problem resulting in numerical issues such as mesh dependency or checkerboarding [12,13]. Overviews of numerical issues can be found for checkerboarding in [14,15] and for mesh dependency in [16–18]. Non-local regularization schemes are commonly used in all of those fields. Penalizing the highest gradient of the internal variable, i.e., gradient-enhanced regularization of the free energy, is well established for this purpose [19–21]. It renders the formerly ill-posed problem into an inherently well-posed one [12]. Due to the insertion of the Laplacian due to the regularization, the emerging partial differential equation is similar to the standard heat or diffusion equation. The number of unknowns increases because not only the displacements have to be evaluated, but also the equations for the additional internal variables. Consequently, the numerical cost increases [12,13]. To ensure increasing complexity of the models which describe the underlying mechanisms, it is essential to develop new numerical treatments, i.e., spatial discretizations that can reduce the time required for simulations while also minimizing the amount of memory space.

In previous publications, an efficient numerical treatment, the so called Neighbored Element Method (NEM), was developed and applied to damage models [12,13,22] and topology optimization [20,23,24]. It was adapted from meshless methods, such as the Diffuse Element Method (DEM), where the least-square approximation is implemented into a Galerkin framework, first introduced by [25–27] and more refined into the element-free Galerkin (EFG) by [28]. Both, small strain and finite strain topology optimization was performed using the NEM formulation. The NEM decouples the calculation of displacements and internal variables and solves the system of differential equations with an operator split in a fully explicitly staggered approach. The displacements are calculated in a standard Finite Element Method (FEM) formulation [29] or similar numerical schemes as for example the Virtual Element Method [30]. The displacements are then used to solve the strong form of the evolution equation for the internal variables. For the aforementioned regularization, the Laplacian of the internal variable needs to be computed. For structured meshes, it is obtained by using a Finite Difference Method (FDM). For unstructured meshes, an inverted Taylor series expansion of the order two is utilized where the first and second derivatives (mixed and unmixed) become the unknowns of a linear system of equations. This generalized Finite Difference Method for arbitrary meshes was presented in [31–33]. To avoid a singular coefficient matrix, a weighted least-square (WLS) is applied where more than the necessary nine neighbors in 3D (five in 2D) are taken into account. Application of the Taylor series expansion with the weighted least-square approach in FEM context can be found in [26,34]. For further information, the reader is referred to [23].

The NEM is an efficient and robust numerical treatment. It conserves the size of the global system of equation, or the primal unknowns, i.e., the displacements for the application of FEM, without internal variables. The update of the internal variable(s) is computed separately, i.e., in a staggered manner, and has only minimal extra computational costs. The goal of this article is to compare the NEM with a standard FEM formulation. For this, the commonly used ANSYS[®] Academic Research Mechanical, Release 2022R1 software has been chosen. In principle, every differential equation with a spatial gradient could have been used as a reference. Here, the built-in elements for thermo-mechanically and chemo-thermo-mechanically coupled materials serve as an example. These built-in elements are chosen, since the ANSYS[®] code for those differential equations is efficiently implemented.

The article is structured as follows. The methodology and mathematical derivation for the approximation of the Laplacian as well as the solution scheme is described in Section 2. Section 3 presents the models for accuracy and efficiency comparison. Numerical results are shown in Section 4, with the comparison of accuracy in Section 4.1 and efficiency in Section 4.2. In Section 4.3, the influence of the numerical parameter is investigated. Finally, in Section 5, conclusions are drawn and possible future work is mentioned.

2. Methodology

In order to study the coupled behavior of physical problems, it is necessary to solve the governing equations. In the majority of mechanical problems, the balance of linear momentum Eq. (1)

$$\text{div } \boldsymbol{\sigma} + \mathbf{b} = \mathbf{0}. \quad (1)$$

must be solved. The evolution of the field variable, that is to say the internal variable or additional degrees of freedom (DOF), e.g., temperature or diffusion, is described in the following manner

$$\mathbf{P} \cdot \dot{\mathbf{v}} = \mathbf{P} \cdot \frac{\partial \mathbf{v}}{\partial t} = \nabla \cdot (\mathbf{Z} \cdot \nabla \mathbf{v}) + \mathbf{Y} \cdot \nabla \mathbf{v} + \mathbf{X} \cdot \mathbf{v} - c. \quad (2)$$

Here, $\mathbf{v} \in \mathbb{R}^l$ with the number of field variables l , is a vector collecting all matrices field variables with the coefficient tensors $\mathbf{P} \in \mathbb{R}^{l \times l}$, $\mathbf{X} \in \mathbb{R}^{l \times l}$, $\mathbf{Y} \in \mathbb{R}^{l \times m \times l}$, $\mathbf{Z} \in \mathbb{R}^{m \times l \times m \times l}$ and $c \in \mathbb{R}^l$ describe the right hand side. The number of dimensions is denoted by m . The equation for the field variables is influenced by diffusion (Laplace term), advection (gradient term) and intrinsic behavior (linear term) which are in principle mutually coupled via the coefficient matrices.

2.1. NEM-scheme

The general procedure of NEM is shown in Fig. 1 where the squared rectangles describe the calculation with FEM and the rounded rectangles the calculation of the generalized FDM, i.e., the inverse Taylor series expansion. The inverted Taylor series expansion of order two is applied, wherein the first and second derivatives are treated as the unknowns of a linear system of equations. It is further explained in Section 2.2. Prior to solving a specific boundary value problem, the coefficient matrix $\mathbf{D}$ has to be calculated once for a given spatial distribution of the neighbored points, i.e., once for a given mesh if no remeshing is employed. The coefficient matrix $\mathbf{D}$ contains information about relative positioning of neighboring points. More elaboration on its derivation is provided in Section 2.2. A change in initial conditions does not result in a recalculation of the coefficient matrix $\mathbf{D}$. Except for the first time step, v is determined based on the current stress σ_t . The resulting right-hand side of Eq. (9) can be used to calculate the gradient or Laplacian of the internal variables. In the last step, the stresses are updated through a Newton–Raphson-iteration step. After converging, we proceed to the subsequent time step.

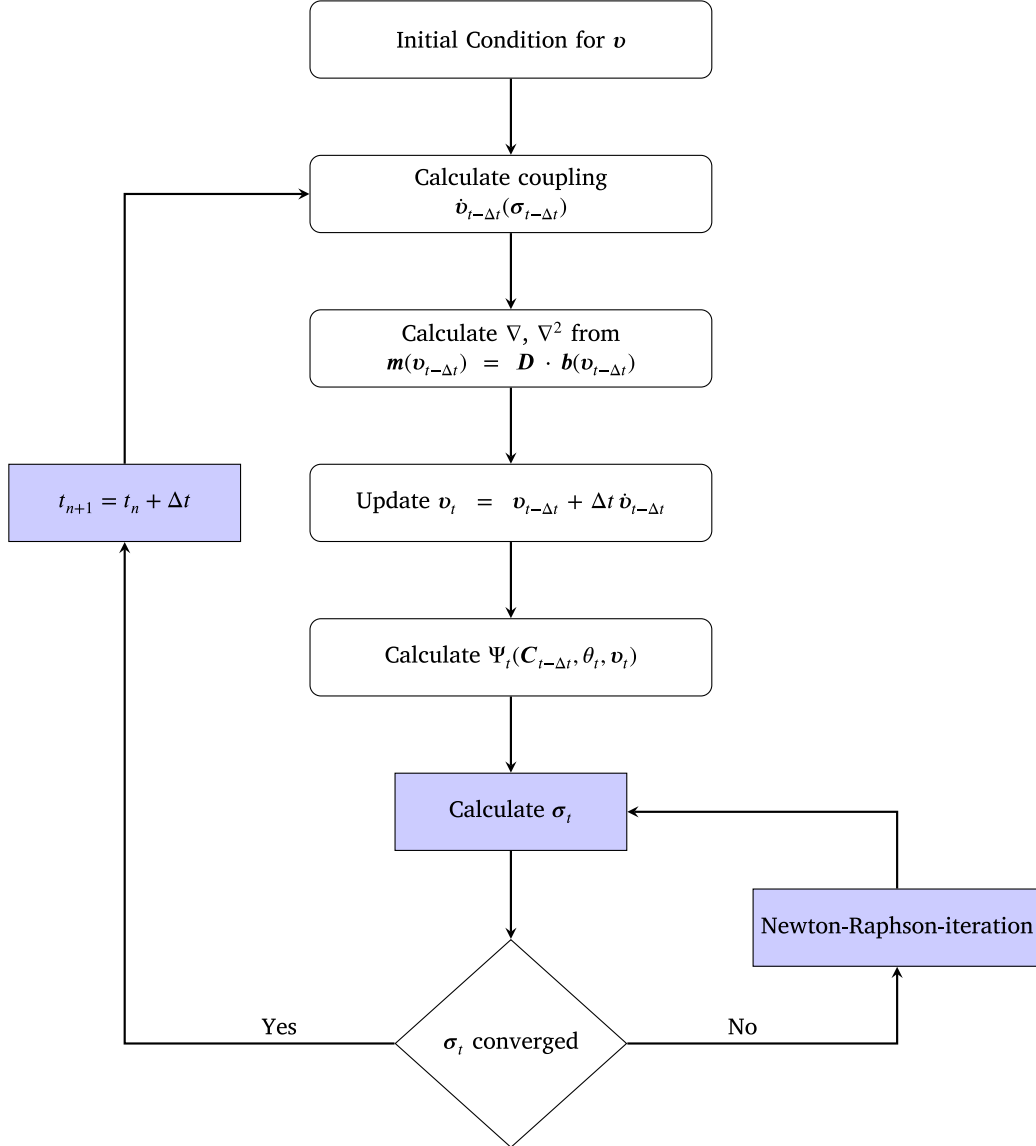


Fig. 1. Scheme of NEM.

For a stable simulation, the step size Δt has to be sufficiently small, i.e., it has to comply to the Neumann-stability criterion

$$\frac{h^2}{2\Delta t} \geq \beta \quad (3)$$

where h is the mesh size and β depends of the smallest (or largest) eigenvalue of the coefficient matrix $\mathbf{Z}$. In the case of heat conduction Eq. (19), β is called thermal diffusivity, as being explained in Section 2.3. In order to achieve a high degree of spatial

accuracy, the time step size Δt must be relatively small. In certain applications that employ explicit time integration, such as damage models, crash simulation or optimization, the use of small time step sizes is already a necessity. Consequently, the reduction of computation times is significant and outweighs any potential increase of the time steps. However, an implicit method can also be employed [20]. Depending on the diffusion coefficient, the implicit method may be overall less numerically efficient since the calculation time scales linearly for the NEM with respect to the number of data points, but quadratically with the size of the diffusion coefficient. In implicit methods, the effort usually scales cubic with the number of degrees of freedom, but is independent of the diffusion coefficient.

2.2. Numerical treatment - approximation of the Laplacian

The proposed scheme builds upon the procedure of the inversion of a Taylor series expansion and combination with the weighted least-square method (WLS) to calculate function values and spatial derivatives for arbitrary point clouds [12,13,20,23,35] and to compute the Laplace operator. The methods presented in previous studies, such as those referenced in [12,35], are subject to limitations. This is due to the difficulty, or even impossibility, of identifying suitable neighbor points in each cardinal direction which is dependent on the selected element type for spatial discretization or the finite element mesh. An inappropriate selection of neighbor points may result in an unsuccessful calculation. This flaw can be resolved by applying more neighbors to the Taylor series and therefore derive an over-determined equation system. This equation system can be solved with a weighted least-square approach. The method is presented in [23] and recalled here for convenience. The Taylor series is defined by a reference point $\mathbf{x}_0 = (x_0, y_0, z_0)^T \in \Omega \setminus \partial\Omega$ and an arbitrary neighbored point $\mathbf{x}_i = (x_0 + \Delta x_i, y_0 + \Delta y_i, z_0 + \Delta z_i)^T$ in a volume Ω . To compute the derivatives of the internal variable v , at least nine neighbored points are required due to the spatial coordinates involved resulting in three first spatial derivative, three mixed second derivatives, and three unixed derivatives. Therefore, nine derivatives need to be found in total. Applying the Taylor series to n unique neighbors yield the equation system represented as matrix form

$$\underbrace{\begin{bmatrix} \frac{1}{2}\Delta x_1^2 & \frac{1}{2}\Delta y_1^2 & \frac{1}{2}\Delta z_1^2 & \Delta x_1\Delta y_1 & \Delta x_1\Delta z_1 & \Delta y_1\Delta z_1 & \Delta x_1 & \Delta y_1 & \Delta z_1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{1}{2}\Delta x_n^2 & \frac{1}{2}\Delta y_n^2 & \frac{1}{2}\Delta z_n^2 & \Delta x_n\Delta y_n & \Delta x_n\Delta z_n & \Delta y_n\Delta z_n & \Delta x_n & \Delta y_n & \Delta z_n \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} v_{xx} \\ v_{yy} \\ v_{zz} \\ v_{xy} \\ v_{xz} \\ v_{yz} \\ v_x \\ v_y \\ v_z \end{bmatrix}}_{\mathbf{m}} = \underbrace{\begin{bmatrix} v_1 - v_0 \\ \vdots \\ v_n - v_0 \end{bmatrix}}_{\mathbf{b}} \quad (4)$$

with $\mathbf{A}$ including the information about the distance between the reference point and the neighboring points, $\mathbf{m}$ containing the unknowns of the problem, i.e., the numerical values of the derivatives. The vector $\mathbf{b}$ contains the information about the values for the internal variables within the data points of the reference point and its neighbors. An amount of $n > 9$ neighbors leads to an over-determined system with more equations than unknown variables. However, an approximate result can still be obtained. Previous studies, such as [22,35], have adopted a right-inverse technique for this aim. To obtain more accurate results, it is possible to include weighting the distances between the reference point and neighbors. An approximation of the result is to minimize the residual which is the total of the quadratic differences between $\mathbf{A} \cdot \mathbf{m}$ and $\mathbf{b}$. The weighted least-square (WLS) problem defined this way reads [36]

$$\min_{\mathbf{m}} \|\mathbf{W}_D^{\frac{1}{2}} \cdot (\mathbf{A} \cdot \mathbf{m} - \mathbf{b})\|_2^2 = \min_{\mathbf{m}} ((\mathbf{A} \cdot \mathbf{m} - \mathbf{b})^T \cdot \mathbf{W}_D \cdot (\mathbf{A} \cdot \mathbf{m} - \mathbf{b})). \quad (5)$$

The weighting matrix $\mathbf{W}$ is a diagonal matrix with the entries $w(d)$

$$w(d) = \exp\left(-\frac{1}{2} \left(\frac{4d}{\beta^*}\right)^2\right) \quad (6)$$

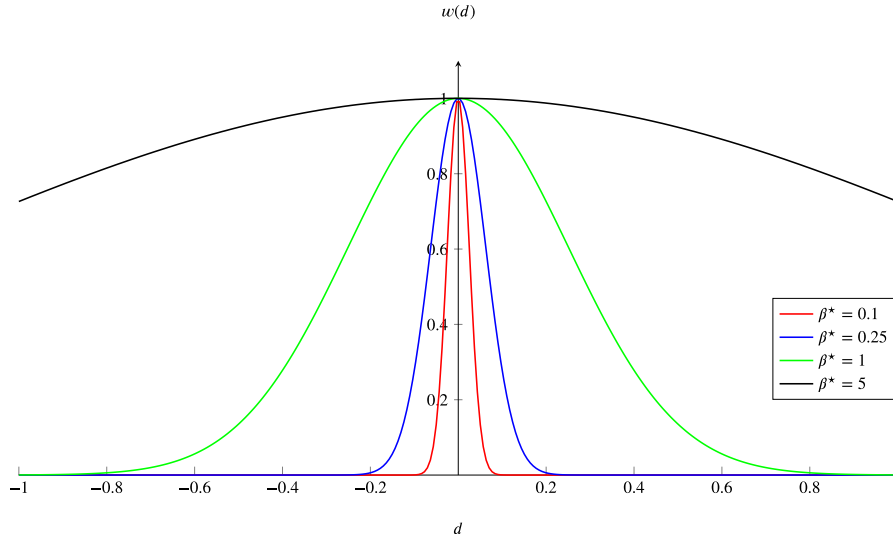
where β^* is defined through the user. In Fig. 2, the influence of β^* on the weighting function is shown. The smaller β^* , the narrower the weighting function gets. If β^* is too small, the equation system becomes singular. For a too large β^* , all neighbors are weighted approximately equally. An investigation of the impact of this numerical parameter on the numerical result and the error is presented in Section 4.3.

The Euclidean distance can be expressed as the following equation, where d_i is the distance between neighboring point i and the reference point $\mathbf{x}_0$

$$d_i = \|\mathbf{x}_i - \mathbf{x}_0\|_2. \quad (7)$$

To reach the minimum solution in Eq. (5), all initial partial derivatives with respect to the function values of all neighbored points must vanish. This results in a linear system

$$\mathbf{0} \stackrel{!}{=} \nabla_{\mathbf{m}} \|\mathbf{W}_D^{\frac{1}{2}} \cdot (\mathbf{A} \cdot \mathbf{m} - \mathbf{b})\|_2^2 = 2 \cdot (\mathbf{A}^T \cdot \mathbf{W}_D \cdot \mathbf{A} \cdot \mathbf{m} - \mathbf{A}^T \cdot \mathbf{W}_D \cdot \mathbf{b}). \quad (8)$$

Fig. 2. Influence of β^* in the exponential weighting function.

and

$$\mathbf{m} = \underbrace{(\mathbf{A}^T \cdot \mathbf{W}_D \cdot \mathbf{A})^{-1} \cdot \mathbf{A}^T \cdot \mathbf{W}_D \cdot \mathbf{b}}_{=: \mathbf{D}}. \quad (9)$$

can be calculated with the coefficient matrix $\mathbf{D}$. The last calculation step can be formulated by multiplying a row vector by a column vector as

$$\nabla^2 \mathbf{v} := \frac{\partial^2 \mathbf{v}}{\partial x^2} + \frac{\partial^2 \mathbf{v}}{\partial y^2} + \frac{\partial^2 \mathbf{v}}{\partial z^2} \approx \sum_{i=1}^3 m_i = \sum_{i=1}^3 \sum_{j=1}^n D_{ij} \cdot b_j. \quad (10)$$

The homogeneous Neumann boundary condition for boundary points $\mathbf{x} \in \partial\Omega$ in the form

$$\frac{\partial \mathbf{v}}{\partial x} n_x + \frac{\partial \mathbf{v}}{\partial y} n_y + \frac{\partial \mathbf{v}}{\partial z} n_z = 0 \quad (11)$$

with n_x , n_y and n_z describing the components of the normal vector $\mathbf{n}$ can be included by adding an additional last row to the matrix $\mathbf{A}$. The adjusted linear system at $\mathbf{x} \in \partial\Omega$ is denoted with $\tilde{\mathbf{A}}$ and reads

$$\underbrace{\begin{bmatrix} \frac{1}{2} \Delta x_1^2 & \frac{1}{2} \Delta y_1^2 & \frac{1}{2} \Delta z_1^2 & \Delta x_1 \Delta y_1 & \Delta x_1 \Delta z_1 & \Delta y_1 \Delta z_1 & \Delta x_1 & \Delta y_1 & \Delta z_1 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ \frac{1}{2} \Delta x_n^2 & \frac{1}{2} \Delta y_n^2 & \frac{1}{2} \Delta z_n^2 & \Delta x_n \Delta y_n & \Delta x_n \Delta z_n & \Delta y_n \Delta z_n & \Delta x_n & \Delta y_n & \Delta z_n \\ 0 & 0 & 0 & 0 & 0 & 0 & n_x & n_y & n_z \end{bmatrix}}_{\tilde{\mathbf{A}}} \underbrace{\begin{bmatrix} v_{xx} \\ v_{yy} \\ v_{zz} \\ v_{xy} \\ v_{xz} \\ v_{yz} \\ v_x \\ v_y \\ v_z \end{bmatrix}}_{\tilde{\mathbf{m}}} = \underbrace{\begin{bmatrix} v_1 - v_0 \\ \vdots \\ v_n - v_0 \\ 0 \end{bmatrix}}_{\tilde{\mathbf{b}}}. \quad (12)$$

As before, we consider more neighbored points and use the WLS algorithm to approximate the solution. Finally, we obtain the formulation

$$\tilde{\mathbf{m}} = \underbrace{(\tilde{\mathbf{A}}^T \cdot \tilde{\mathbf{W}} \cdot \tilde{\mathbf{A}})^{-1} \cdot \tilde{\mathbf{A}}^T \cdot \tilde{\mathbf{W}} \cdot \tilde{\mathbf{b}}}_{=: \tilde{\mathbf{D}}}. \quad (13)$$

In the three dimensional case, the inverse of $(\tilde{\mathbf{A}}^T \cdot \tilde{\mathbf{W}} \cdot \tilde{\mathbf{A}})$ is a 9x9 matrix and independent of the number of neighboring points. The inverse in $\tilde{\mathbf{D}}$ is thus not computationally expensive to be computed. Furthermore, $\tilde{\mathbf{m}}$ remains constant when no mesh adaptivity is employed. However, the method is still fast even when mesh adaptivity is employed, cf. [22,35]. The $\mathbf{W}$ -matrix for the whole

system therefore is defined as the diagonal-matrix

$$\tilde{\mathbf{W}} = \begin{bmatrix} w(d_1) & 0 & 0 & \dots & 0 \\ 0 & w(d_2) & 0 & \dots & 0 \\ 0 & 0 & w(d_3) & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \tilde{w} \end{bmatrix} \quad (14)$$

with the matrix $\tilde{\mathbf{D}}$ containing the WLS information for all neighbored points. The weighting for the points at the boundary is denoted with $\tilde{w}$. It should be noted here that to solve Eq. (9), the inverse of a matrix has to be calculated to solve the Eq. (9). The matrix is invertible if $\mathbf{A}$ has full column rank which is the case if less than $(n - 2)$ of the n nearest neighbored points lie on a straight line together with the reference point [23].

2.3. Chemo-thermo-mechanically coupled materials as benchmark problem

The stresses for chemo-thermo-mechanically coupled problems can be calculated by

$$\boldsymbol{\sigma} = \mathbb{E} : \boldsymbol{\epsilon}^{\text{el}} = \mathbb{E} : (\boldsymbol{\epsilon}^{\text{tot}} - \boldsymbol{\epsilon}^{\text{p}}) = \mathbb{E} : (\boldsymbol{\epsilon}^{\text{tot}} - \boldsymbol{\epsilon}^{\text{th}} - \boldsymbol{\epsilon}^{\text{diff}}). \quad (15)$$

Eq. (15) shows the definition of the stress tensor $\boldsymbol{\sigma}$, the stiffness tensor $\mathbb{E}$ and the strain tensor $\boldsymbol{\epsilon}^{\text{el}}$. The strain tensor is the difference between the total strain $\boldsymbol{\epsilon}^{\text{tot}}$ and field variable strain $\boldsymbol{\epsilon}^{\text{p}}$. In this presented work, the field variable strain is the sum of the thermal $\boldsymbol{\epsilon}^{\text{th}}$ and diffusion strain $\boldsymbol{\epsilon}^{\text{diff}}$. The thermal strain is defined as

$$\boldsymbol{\epsilon}^{\text{th}} = \alpha_{\text{th}} \Delta\theta \mathbf{I} \quad (16)$$

with the isotropic coefficient of thermal expansion α_{th} and the identity matrix $\mathbf{I}$. The diffusion strain results from

$$\boldsymbol{\epsilon}^{\text{diff}} = \alpha_{\text{diff}} \Delta c \mathbf{I} \quad (17)$$

with the isotropic coefficient of diffusion expansion α_{diff} . Here, we specify the field equation in Eq. (2). To this end, we introduce as vector with additional field variables

$$\mathbf{v} := \{\theta, c\} \quad (18)$$

whereas θ and c describe the temperature and material concentration, respectively. The coefficient matrices in Eq. (2) are defined as $\mathbf{P} := \begin{bmatrix} \rho c_p & 0 \\ 0 & 1 \end{bmatrix}$, with c_p as the specific heat capacity (at constant pressure) and ρ indicating the density of the material. $\mathbf{X}$ and $\mathbf{Y}$ are zero matrices. The non-zero entries for the $\mathbf{Z}$ matrix are the heat conductivity λ and diffusion coefficient κ . The vector $\mathbf{c} := \begin{bmatrix} Q \\ 0 \end{bmatrix}$ contains the volumetric heat source Q . These specifications of the coefficient matrices result in the well-known heat equation

$$\rho c_p \frac{\partial \theta}{\partial t} - \nabla \cdot (\lambda \nabla \theta) = Q \quad (19)$$

and in the diffusion equation

$$\frac{\partial c}{\partial t} = \kappa \nabla^2 c. \quad (20)$$

In the specific case of heat transfer in an isotropic and homogeneous medium in a three-dimensional space with no heat source ($Q = 0$), Eq. (19) simplifies to

$$\frac{\partial \theta}{\partial t} = \frac{\lambda}{\rho c_p} \nabla^2 \theta. \quad (21)$$

3. Simulation model used for comparison

The ANSYS[®] software is used in this work. This renowned program provides the ability to alter and modify established procedures as needed. This can be accomplished using the APDL ANSYS[®] Parametric Design Language or user subroutines built on the Fortran platform. Here, APDL is employed. It can be simply integrated into the ANSYS[®] user interface and provides all necessary functionalities. Two key perspectives are examined in this analysis. First, a comparison of the results' accuracy is investigated. A quarter-torus model including the thermo-mechanical coupling as an example equation is employed for this purpose. Then, a comparison is made between the NEM implementation and the traditional FEM. The second factor is efficiency. To be more specific, we focus on the resources Random-Access-Memory (RAM) and the processing time. We investigate the performance based on two exemplary coupled problems: the first one of these is the thermo-mechanical coupling, and the second one is the chemo-thermo-mechanical coupling.

Table 1
Material properties steel.

Properties	Symbol	Unit	Value
density	ρ	kg m^{-3}	7850
Young's Modulus	E	GPa	200
specific heat capacity	c_p	$\text{J kg}^{-1} \text{K}^{-1}$	434
heat conductivity	λ	$\text{W m}^{-1} \text{K}^{-1}$	60.5
isotropic coefficient of thermal expansion	α_{th}	K^{-1}	$1.2 \cdot 10^{-5}$

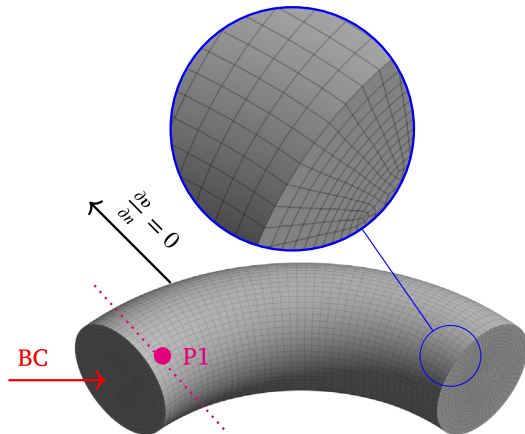
3.1. Model for the evaluation of the accuracy of the NEM

To evaluate whether the NEM is an accurate scheme for the numerical solution of the thermo-mechanical problems, which serves as an example equation system, a comparison with standard FEM is necessary. A quarter torus is selected as the reference geometry due to its inherent complexity, characterized by the coexistence of two distinct curvatures. The inner radius is $r = 0.5 \text{ mm}$ and the outer radius is $R = 2 \text{ mm}$. The whole model with boundary conditions and mesh are shown in Fig. 3(a) and Fig. 3(b), respectively. The material properties chosen are the standard ANSYS® steel properties. They are listed in Table 1.

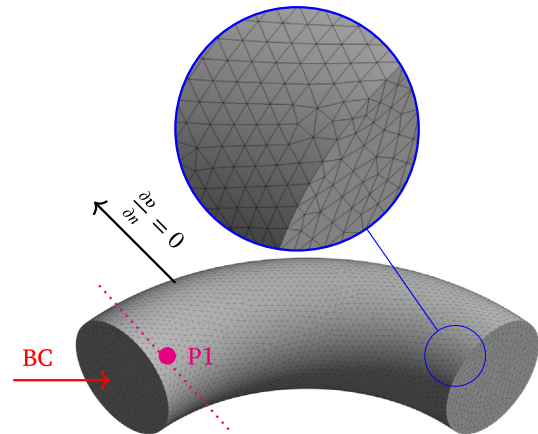
The calculation is a coupled field simulation comprising three DOF's for displacements and one additional DOF for the additional field variable, which is employed here as an illustrative example of temperature. It is assumed that no heat flow can occur over the boundary surfaces, i.e., perfect isolation. Hence, the Neumann boundaries conditions Eq. (11) are implemented in NEM to ensure perfect isolation. Furthermore, the red arrow labeled with BC indicates the plane surface for the non-zero boundary condition. The temperature of the surface remains constant at $\theta_0 = 1^\circ\text{C}$ throughout the duration of the simulation. All nodes on the surface BC are defined as fixed supports of displacements. The initial temperature of all other nodes is given by $\theta = 0^\circ\text{C}$. Furthermore, the time step size, $\Delta t = 2 \times 10^{-5} \text{ s}$, is employed. The final time point, $t = 0.1 \text{ s}$, is reached after 5000 time steps.

The meshing algorithm from ANSYS® software with standard elements was applied. Both, regular and irregular meshes were investigated in this model. The regular mesh is created by hexahedral elements *SOLID185* with linear shape functions. The nominal edge length of these elements is $h = 0.05 \text{ mm}$ which results in 34965 elements and 37260 nodes, respectively. To generate the irregular mesh, tetrahedral elements *SOLID185* with linear shape function are applied. Furthermore, additional information regarding the specific element type can be found in the ANSYS® manual [37]. This results in 68819 elements and 14728 nodes, respectively. In addition, the Neumann-stability criterium in Eq. (3) is satisfied by the given material properties, the time step size and the mesh size.

The same meshes are used in the FEM and NEM implementation in ANSYS®. This allows a direct comparison between the nodal values. The ANSYS® simulation serves as benchmark for these results. The point P1 in Fig. 3 indicates the node position at which the values of the results will be compared. This node is located inside of the body, surrounded by neighboring nodes. The dotted magenta line indicates the position of cross-section surface for further visual comparison.



(a) Torus model with hexahedral elements, measurement points and outer radius.



(b) Torus model with tetrahedral elements, boundary conditions and inner radius.

Fig. 3. Torus meshes.

3.2. Model for the evaluation of the efficiency of the NEM

To investigate the efficiency of the NEM, the thermo-mechanical and chemo-thermo-mechanical problem were solved. The results were compared to results obtained with built-in elements in ANSYS® and evaluated using two criteria. The first one is the comparison

of the main CPU thread time required. It indicates, how long it takes for the main thread CPU time to solve the boundary value problem. The second criterion is the required RAM. Both information are reported in the ANSYS[®] solver output file. The comparison was carried out on a computer cluster system with 10 threads and 500 GB RAM. To identify the scattering of the CPU time and RAM on the computer system, the model was calculated 20 times with 105300 nodes. It results in an average calculation time of the main thread CPU of 6323.1 s with a standard deviation of 302.6 s. This corresponds to 5% and is within reasonable limits. The investigation of the standard derivation for the RAM shows an even clearer picture. The differences in required RAM and average calculation time of the main thread CPU between the repeated program executions are negligible.

In order to facilitate a meaningful comparison, a cube was selected as the model against which the efficiency is to be evaluated. The cube is a simple structure to not overextend the computers resources. The edge length of this cube is 1 mm as shown in Fig. 4. Moreover, the material properties of the cube are identical to those of the torus, with the additional diffusivity coefficients $\kappa = 17.75 \text{ m}^2 \text{ s}^{-1}$ and coefficient of diffusion expansion $\alpha_{\text{diff}} = 1.2 \cdot 10^{-5} \text{ m}^3 \text{ mol}^{-1}$ for the concentration.

A coupled 3D system of displacements and additional DOF's is used for the efficiency comparisons. In this case, the additional DOF's are temperature and diffusion. The required solving algorithm for coupled field problems in ANSYS[®] is restricted to the direct and monolithic solver [38]. However the NEM does not necessitate this type of solver as it can be calculated as a decoupled problem. Consequently, the iterative solver can be employed. In this work, the three displacements DOF's are calculated with FEM, while the remaining DOF's are calculated using the inverse Taylor series. Therefore, there are five different calculation models. An overview of these models is shown in Table 2.

Table 2
Overview of the model name and the degrees of freedom.

Method	DOF per node	Solver	Name
FEM	4	Direct	ANSYS4
FEM	5	Direct	ANSYS5
NEM	4	Direct	NEM3+1
NEM	5	Direct	NEM3+2
NEM	5	Iterative	NEM3+2-Iter

The boundary conditions are similar to the ones of the torus. On all surfaces, Neumann boundary condition prevent the flow over the surfaces. In an area in the middle of the model with the dimension of 0.1 mm in each direction, the initial conditions of $\theta_0 = 1^\circ \text{C}$ are applied with the initial temperature of $\theta_0 = 0^\circ \text{C}$ in the remainder of the body. The initial and boundary conditions for diffusion are identical to those for temperature. As illustrated in Fig. 4, the blue spots indicate the locations where mechanical fixed support of displacements has been employed. Time discretization is conducted over 40 time steps, with a constant time step size dependent on the mesh size. This approach is employed to fulfill Eq. (3).

Here, the standard meshing algorithm from ANSYS[®] with *SOLID186* and *SOLID226* elements is used. A regular mesh with 20 nodes hexahedral elements is applied. The element sizes of each mesh range from 0.1 mm to 0.018 mm which corresponds to 4961 up to 731025 nodes. An overview of the different meshes is shown in Table 3.

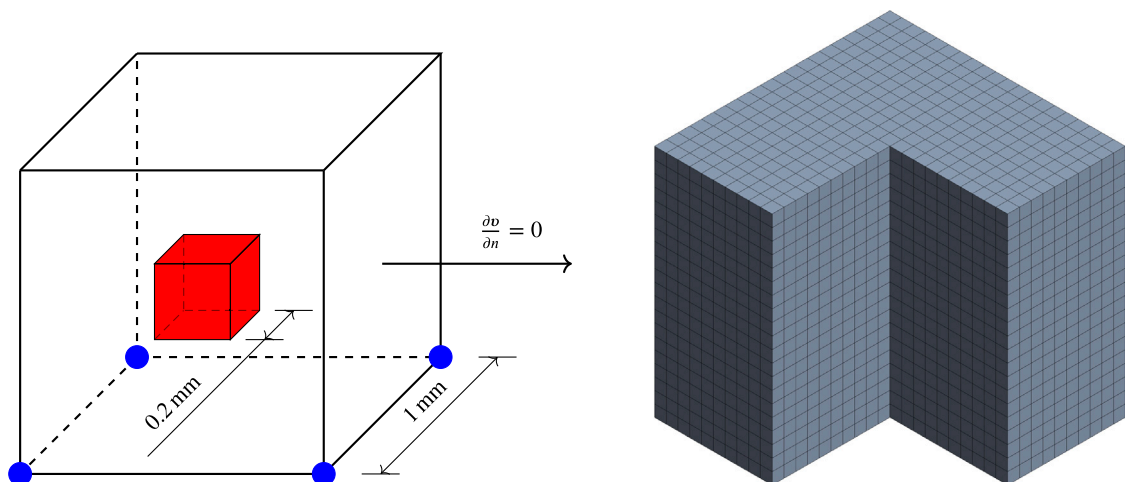


Fig. 4. Cube with initial condition (left side) for an hexahedral mesh element size of 0.05 mm (right side).

Table 3
Cube mesh overview.

Cube nr.	Element size [mm]	Node count [–]
1	0.1	4961
2	0.05	35 721
3	0.035	105 300
4	0.025	270 641
5	0.022	408 665
6	0.02	522 801
7	0.019	621 108
8	0.018	731 025

4. Results

4.1. Evaluation of the accuracy of the NEM

The evolution of the temperature, through the torus with hexahedral elements by using the NEM is demonstrated in Fig. 5. The results are in line with an expected behavior. As conduction through the torus progresses, the temperature increases gradually through the domain. The propagation of the contour front shows that the system tends to reach the steady-state determined by the uniform boundary value of unity. Here, the calculation was carried out until all values have reached a significant difference from the initial temperature which is in this case at about $t = 0.1$ s. As illustrated in Fig. 3(a), the boundary surface labeled *BC* demonstrates zero deformation across all nodes. Conversely, the remaining nodes display a progressive expansion in conjunction with the diffusion of temperature throughout the body which is indicative of the thermoelasticity coupling phenomenon.

As local error measurement, the relative difference between the ANSYS[®] and the NEM solution is computed for each node of the mesh. It is normalized by the ANSYS[®] solution, i.e.

$$\text{err}_k^t = \frac{|\theta_{A,k}^t - \theta_{N,k}^t|}{\theta_{A,k}^t}. \quad (22)$$

The indices *A* and *N* stands for ANSYS[®] and NEM, respectively. The subscript *k* is the node number and the superscript *t* describes the time step.

The results on the cross-section surface for the hexahedral and tetrahedral meshes are presented in Fig. 6. The maximum error value tends to zero over time. The comparison is subject to an error of 0.452 % at $t = 0.005$ s, while at $t = 0.1$ s, the maximum error is reduced to 0.003 %. The tetrahedral mesh also provides a clear visual representation. This suggests that the position of the maximum error is not as localized as it is in the hexahedral mesh. They are distributed in a scattered manner. The maximum error value does not reduce as rapidly as the one observed in the hexahedral mesh. At $t = 0.005$ s, the maximum error is 0.585 % which is approximately 30 % higher than that of the hexahedral mesh. The tetrahedral mesh reaches a maximum error of 0.022 % at $t = 0.1$ s which is seven times higher than the error reported for the hexahedral mesh. Consequently, the hexahedral possesses a smaller relative error than the tetrahedral mesh on this cross section surface.

The investigation of the maximum relative error as well as the average error over the entire torus is presented in Fig. 7. Data is taken every $t = 0.001$ s. The image provides a clear interpretation of the error evolution in time. With ongoing time, the error asymptotically approaches to zero while the diffusion is still in progress. The maximum error of the tetrahedral mesh is higher than the error of the hexahedral mesh. The accuracy of the method is reliant upon the equiangularity of the distribution of the neighboring points around the reference point. An almost equiangular distribution is exhibited by a structured mesh and therefore, the numerical error is smaller for the hexahedral mesh. This behavior was also presented in the previous example, c.f. Fig. 6. Initially, the maximum error exceeds 1.5 % and 5.5 %, respectively. For the hexahedral mesh, the error was reduced to below 1 % at $t = 0.002$ s, while for the tetrahedral mesh, this was achieved at $t = 0.005$ s. The initial average value is below 1 %. It falls below 0.2 % at $t = 0.005$ s for the hexahedral mesh and at $t = 0.003$ s for the tetrahedral mesh.

The convergence of the maximum relative error is characterized by both stagnation and jumps. For instance, at $t = 0.013$ s the tetrahedral mesh exhibits a jump. In each instance of the aforementioned jump, the location of the maximum error shifts from one node to another, resulting in a different denominator in Eq. (22) and, consequently, a sudden alteration in the local error measurement. Conversely, the investigation of the absolute error results in convergence, indicating that the calculation is accurate. Therefore, this jump occurs due to the high change in the basis value for the relative error. In Fig. 7, the error between NEM and ANSYS[®] decreases over time.

The investigation of the spatial discretization in the cross-section surface for the hexahedral mesh is presented in Fig. 8. In order to investigate the convergence of the solution with respect to the mesh size, four different mesh sizes were utilized at $t = 0.005$ s, i.e., 0.02 mm, 0.035 mm, 0.05 mm and 0.065 mm. It can be observed that the solution converges with a finer mesh. The maximum error between the 0.05 mm mesh and the 0.02 mm mesh is approximately half of the former. Furthermore, the mean values are decreased by approximately 37 %.

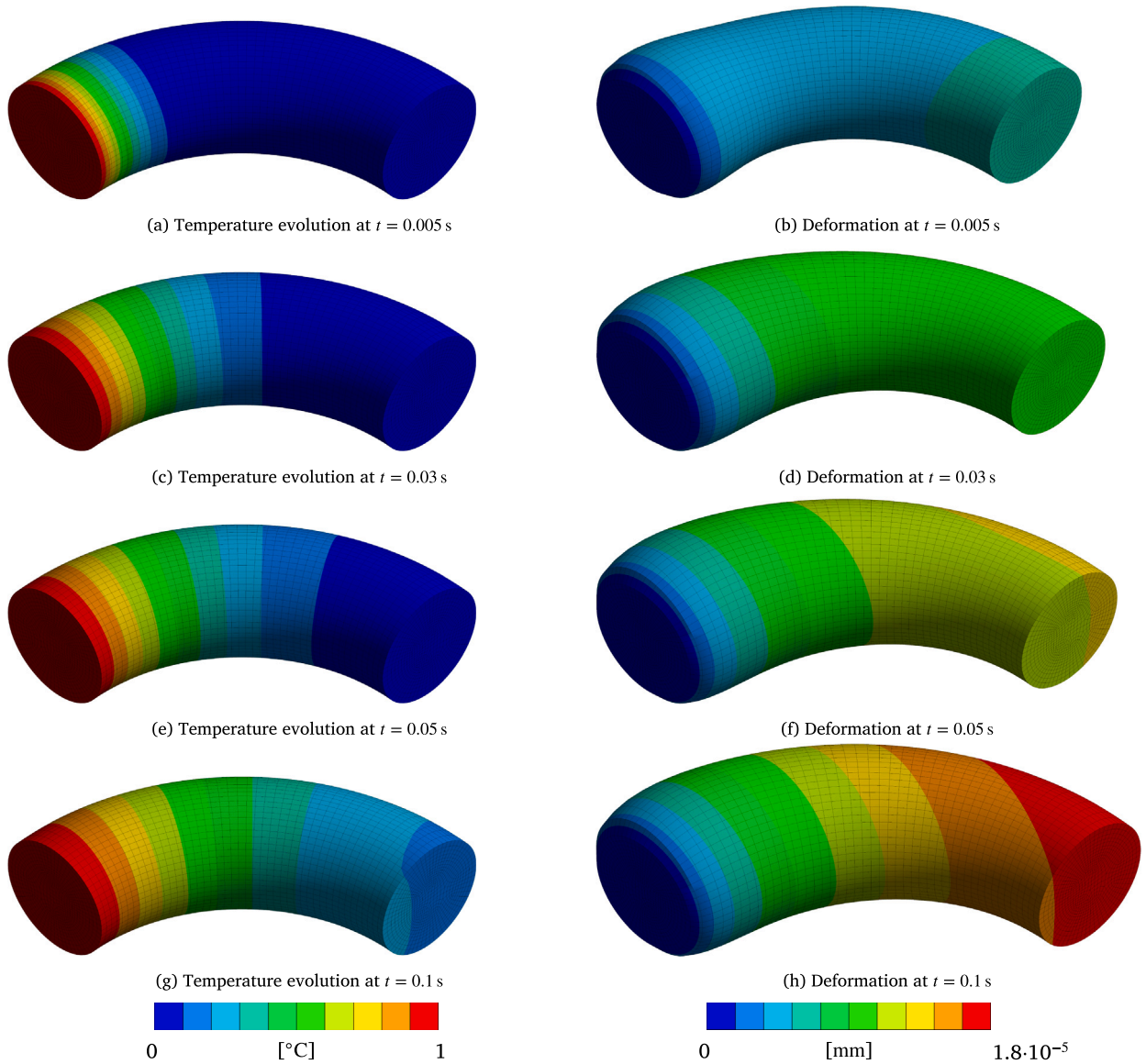
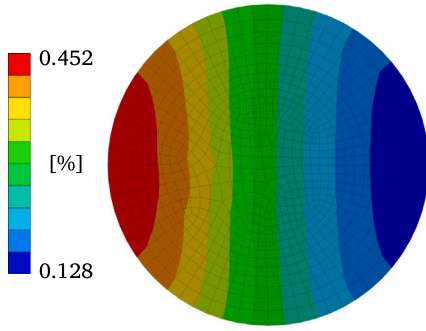
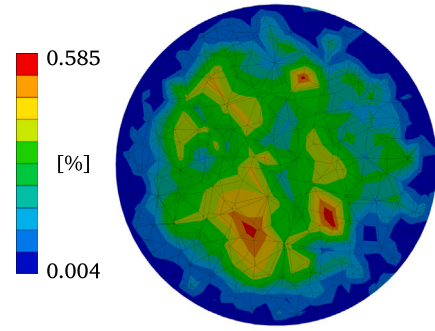
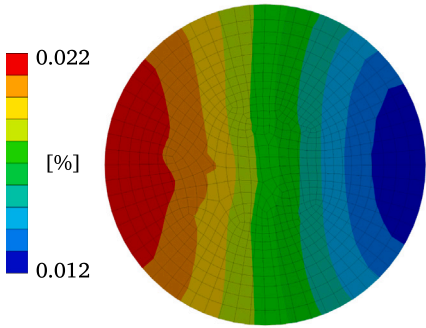
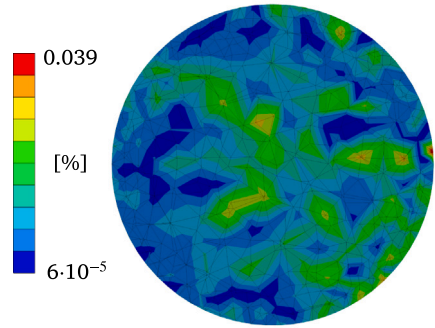
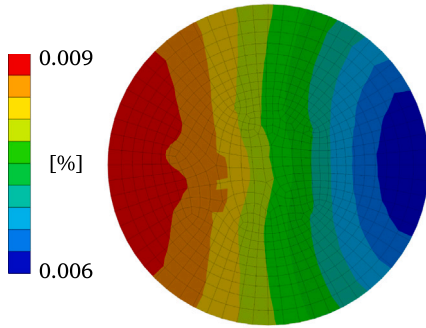
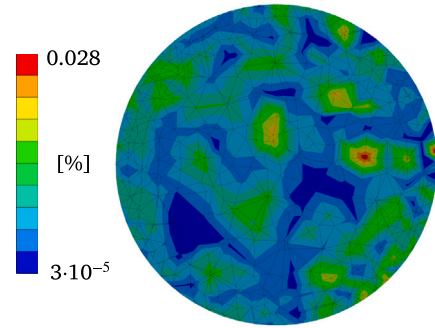
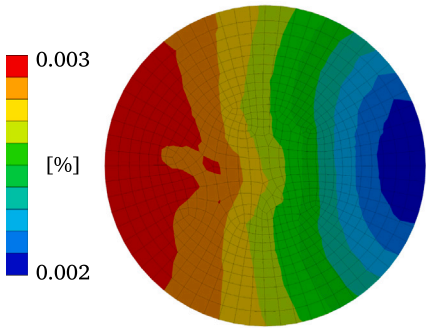
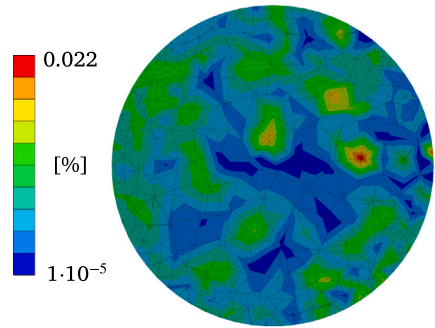


Fig. 5. NEM results for hexahedral elements as diffusion through the torus and deformation at different times, i.e. $t = 0.005$ s to $t = 0.1$ s.

4.2. Evaluation of the efficiency of the NEM

This segment of the study presents a comparison between the two methods in terms of computational time and RAM requirements. Obviously, these requirements depend partly on the implementation of each method and therefore, the conclusions have to be considered carefully. This is particularly true since the implementation of the NEM could be implemented in an even more optimized way, e.g., when implemented via the FORTRAN interface and not via the APDL as done here. The graph in Fig. 9 shows the computation time for the main CPU thread in seconds versus the number of nodes. In all cases, the calculation of the coefficient matrix $\mathbf{D}$ is excluded, as it is not necessary to recalculate it for changes to the initial conditions. Even for the smallest number of nodes 4961, the difference between ANSYS[®] and the NEM is already visible. This difference extends with an increase in the number of nodes. The NEM requires less computation time than the standard Finite Element Method with ANSYS[®].

The difference between NEM3+1 and NEM3+2 is small. There are only few seconds between these calculation. As an example, the calculation time difference is approximately 25s for 270,641 nodes. Therefore, the graphs NEM3+1 —■— and NEM3+2 ···◆··· are overlapping in Fig. 9. The ANSYS5 model with 731025 nodes was too large for computing on the computer cluster system hardware, because the required computation time exceeded the walltime limit. Therefore, a value of 40th was extrapolated from models with the same amount of DOF but fewer time steps. In Fig. 9, it is indicate through the dotted blue line. The largest fully computed

(a) Hex-Mesh, $t = 0.005$ s(b) Tetra-Mesh, $t = 0.005$ s(c) Hex-Mesh, $t = 0.03$ s(d) Tetra-Mesh, $t = 0.03$ s(e) Hex-Mesh, $t = 0.05$ s(f) Tetra-Mesh, $t = 0.05$ s(g) Hex-Mesh, $t = 0.1$ s(h) Tetra-Mesh, $t = 0.1$ sFig. 6. Relative errors err'_k on cross-section surface for different time instances and meshes.

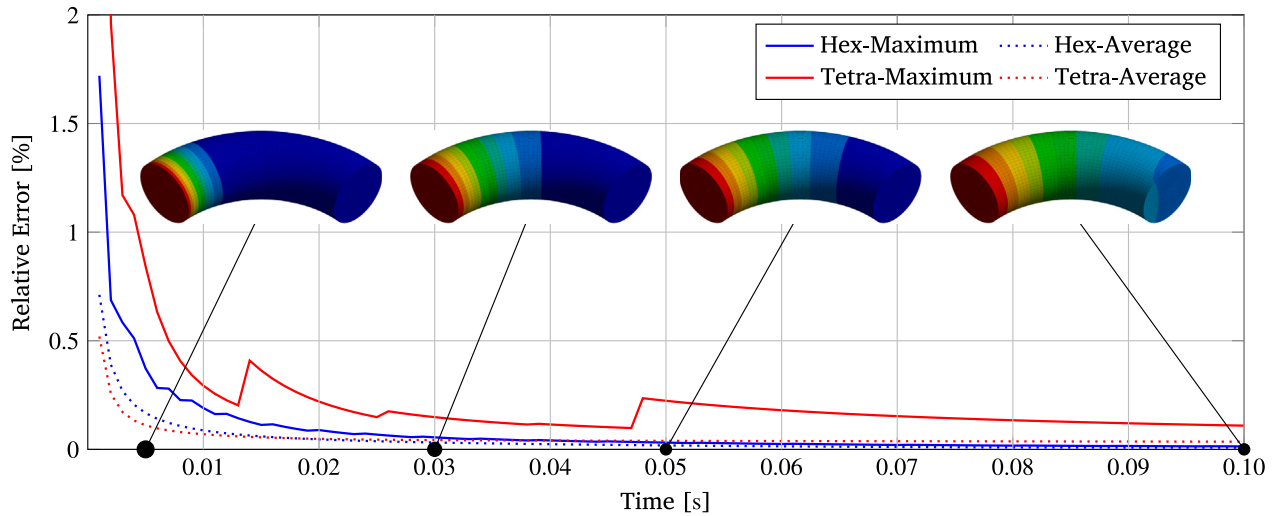


Fig. 7. Maximum relative and average error of the whole torus for hexahedral and tetrahedral elements and the evolution of the temperature diffusion for chosen time (compare Fig. 5).

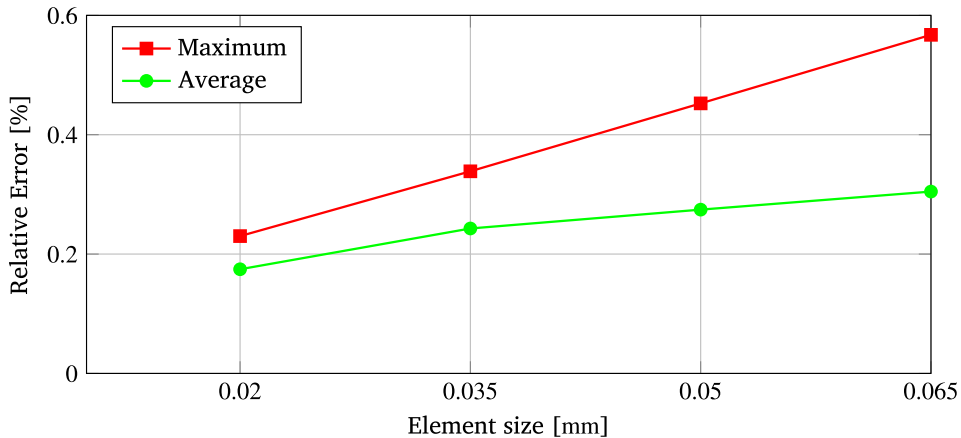


Fig. 8. Relative maximum and average error on the cross-section surface by different hexahedral element sizes by $t = 0.005$ s.

data point, 621108 nodes, was chosen to indicate the time in hours. The NEM3+1 is 5.4 times faster than ANSYS4. NEM3+2 is 9.6 times faster than ANSYS5. Furthermore, with one additional DOF, the NEM3+2 is 5.3 times faster than ANSYS5. As mentioned in Section 3.2, there is no restrictions on the choice of the solver. Therefore, the NEM3+2-Iter was calculated with an iterative solver, resulting in a significant speed increase compared to the method with direct solver. The required computation time for ANSYS5 and NEM5-Iter at 621108 nodes is 88.5 h and 0.57 h, respectively. This corresponds to a 155-fold increase in computation speed.

The comparison for the required RAM are shown in Fig. 10 which also present a clear distinction. The values for ANSYS5 by 731025 nodes are not available, because this model exceed the limit of 500 GB RAM and therefore, no in-core solution is possible. Thus, the data point with 621108 nodes is used to compare the method with respect to RAM.

The difference between NEM4 and NEM5 is small, i.e., for 270641 nodes the difference is 0.15 GB. The comparison of the ANSYS4 and ANSYS5 with NEM3+1 or NEM3+2 gives clear results. The NEM required 3.4 and 5.1 times less RAM than ANSYS4 and ANSYS5, respectively. Moreover, the NEM3+2-Iter only needs 5.1 GB RAM in contrast to ANSYS5 with 394.4 GB. This is a reduction of RAM by a factor of 77.3.

4.3. Influence of the numerical parameter β^*

The investigation of the weighting function as shown in Eq. (6) and the user-selected parameter β^* in the cross-section surface at $t = 0.005$ s is illustrated in Fig. 11. The results of the maximum, average, and local errors at position P1 in percentage by two different counters with 30 and 100 neighbors are presented, respectively. The impact of β^* on the relative error is minimal. The greatest discrepancy between $\beta^* = 0.1$ and $\beta^* = 1$ with 100 neighbors is a reduction of 0.24 % in the maximum relative error in the

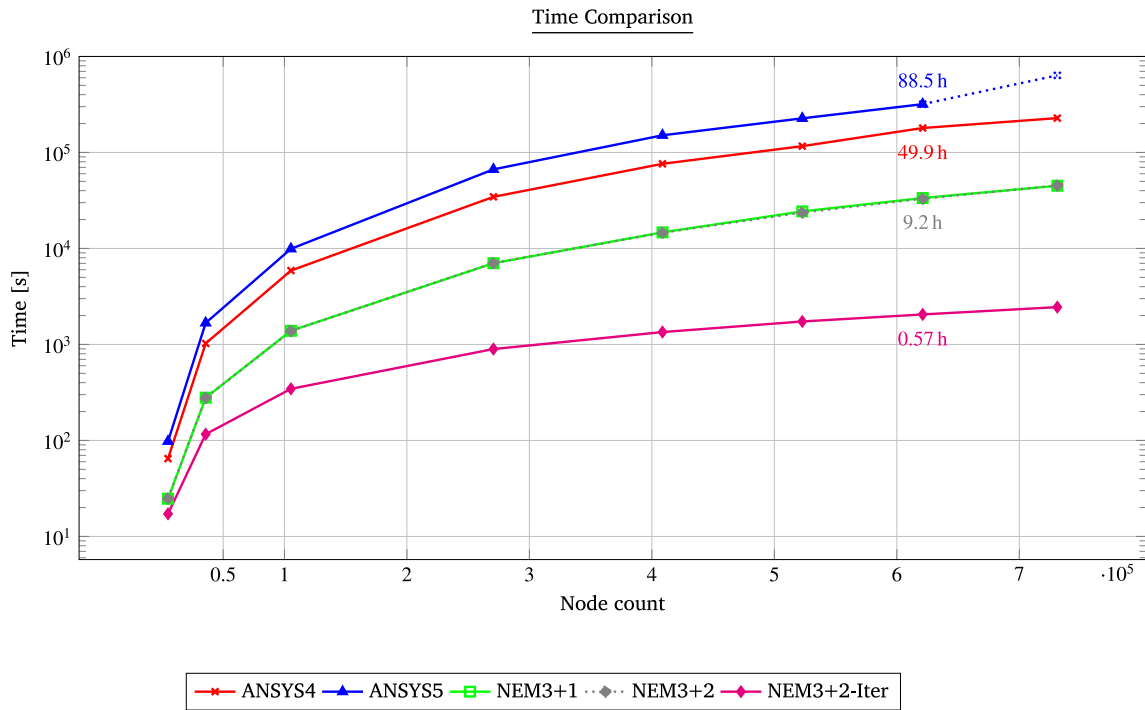


Fig. 9. Time comparison between NEM and FEM.

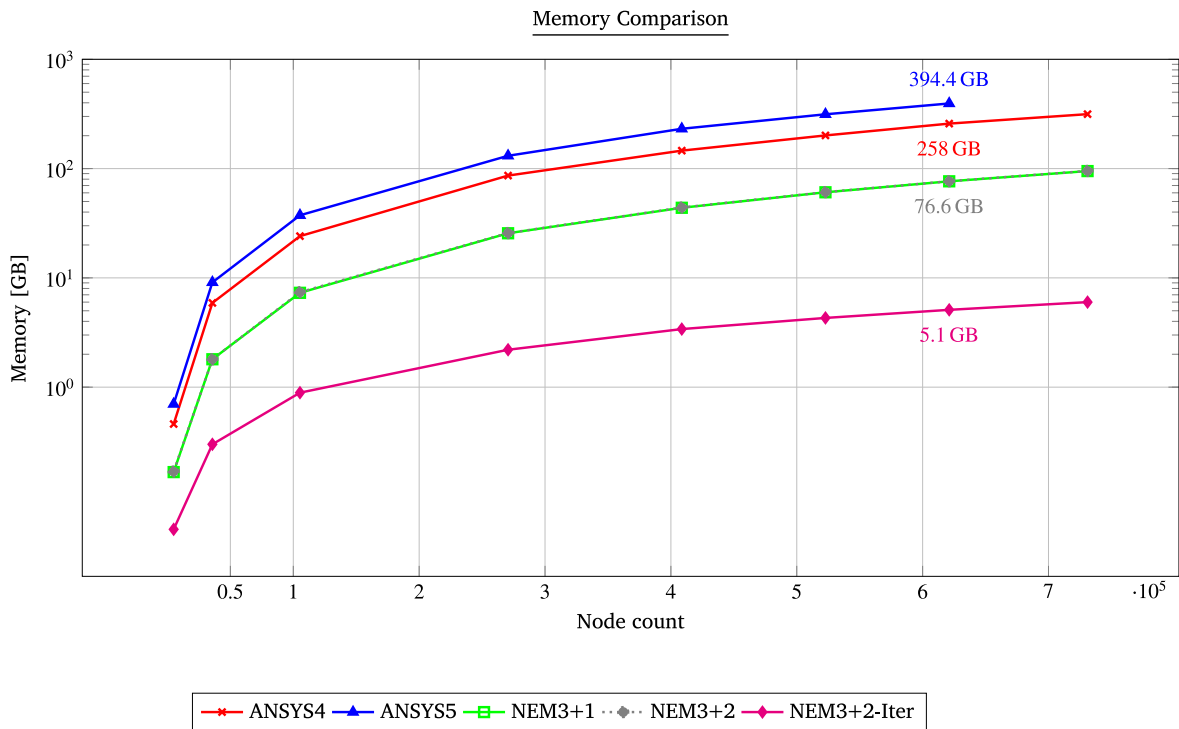


Fig. 10. Memory comparison between NEM and FEM.

cross-section. The average and P1 values demonstrate a relatively constant range of variation, with a maximum difference of 0.26 % and a minimum difference of 0.2 %. These results indicate that the choice of β^* is not a significant factor in this method.

The influence of the parameter $\tilde{w}$ in Eq. (14) is not considered in this work. This parameter should be relatively large in comparison to the number of neighbors. However, no upper boundary for this value was found.

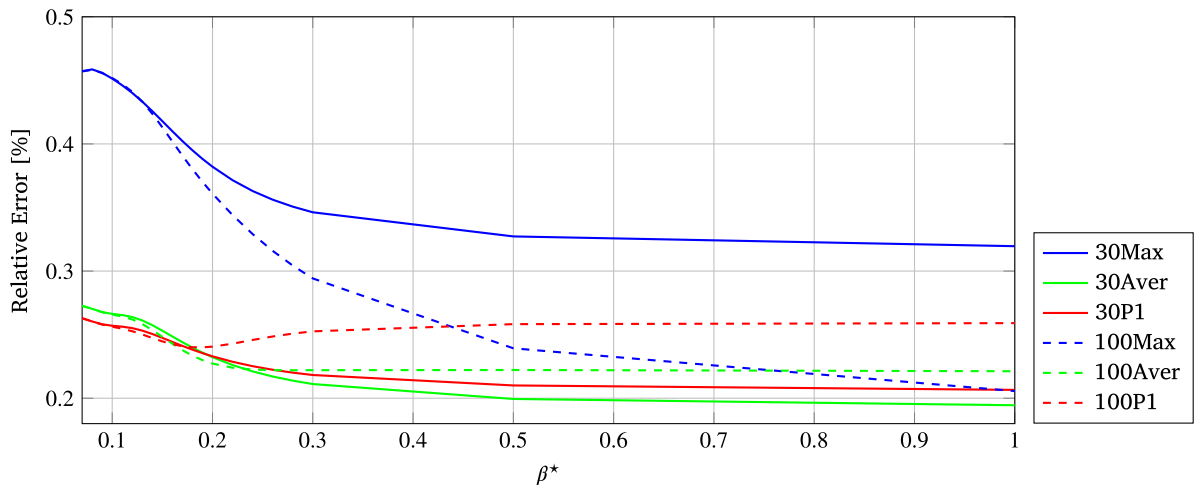


Fig. 11. Influence of β^* in cross-section of hexahedral torus.

5. Conclusion and outlook

This work presents a comparative study between the standard FEM approach and the Neighbored Element Method (NEM) evaluated in ANSYS[®]. The combination of FEM and the inverse Taylor series expansion with the weighted least-square method in NEM was evaluated through the comparison of two distinct models. The accuracy and resource requirements of this method were investigated. The results of the accuracy assessment are presented as a relative error which has been normalized in relation to the ANSYS[®] solution. The results indicate a linear convergence in spatial discretization and a convergent solution in time progression. There is no mesh dependency, and the method is applicable to hexahedral and tetrahedral meshes. The computational resources comparison demonstrates a clear saving of calculation time and RAM by orders of magnitude. The spatial discretization by using the NEM can be different to the FEM which creates more options for resolving coupled problems.

While the results demonstrate that the NEM significantly reduces computation time and memory consumption, it must be considered that this is partly due to the choice of a staggered approach. A comparison with a commercial staggered implementation would be necessary to further quantify the specific advantages of the NEM. This is left for further investigations. In future works, the influence of the value of the weighting function and meshless methods or hybrid meshes, i.e., meshes made from hexahedral as well as tetrahedral elements, can be investigated further. Moreover, the NEM can be applied to a number of additional research topics, including phase transformation, as well as fatigue and biological modeling.

CRediT authorship contribution statement

Tobias Rudolf: Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Felix Klempt:** Writing – original draft, Visualization, Methodology, Investigation. **Hüray İlayda Kök:** Writing – original draft, Visualization, Methodology, Investigation. **Meisam Soleimani:** Writing – review & editing, Supervision. **Dustin Roman Jantos:** Writing – review & editing, Supervision, Methodology. **Philipp Junker:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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